

AAYUSH K. SHARMA

Software Engineer | Developer Relations & Technical Community

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TypeScript-first software engineer who builds AI agent workflows and ships production systems while working directly with end users—debugging their issues, translating field feedback into fixes, and presenting technical work to non-technical audiences. Grew a 15,000-line community platform from 25 to 180+ active members, cutting outages 73% through hands-on support and automation. Built agentic microservices with tool-calling, SQL execution, and retrieval pipelines, and practices AI-native, agent-driven development (Claude Code, Cursor, prompt engineering) daily. B.S. Applied Mathematics.

SKILLS & INTERESTS

Languages: TypeScript, JavaScript, Python, SQL/PSQL, C++, C, Bash, MATLAB

AI-Native/Agentic Development: Claude Code, Claude Design, Cursor, agent workflows & tool-calling, prompt engineering, agentic test-driven development, automation & QA/eval loops

Frameworks/Cloud: Node.js, Express.js, ReactJS, Next.js, FastAPI, PostgreSQL, Azure (IaC/Bicep, Foundry), Digital Ocean, Git, GitHub Actions, Docker, REST APIs, CI/CD

Community & Communication: technical community management, developer support & debugging, content creation, live demos/presentations, cross-functional & customer-facing communication

PROFESSIONAL EXPERIENCE

AI Integrations Engineer

Dallas, TX

Valiant Residential | Business Intelligence

May–Aug 2025 | Jun 2026–Present

- Architected and delivered two agentic microservices with independent tool-calling, SQL execution, and retrieval pipelines during a Summer 2025 internship—hands-on agent-workflow and automation engineering—building a persistent harness layer that standardized schema contracts and query guardrails across heterogeneous SQL backends.
- Now driving end-to-end delivery of a full-stack legal contract analysis platform (React/TypeScript + FastAPI) that lets branch-office staff upload contracts for automated compliance review—gathering requirements directly from users, debugging their issues, and shipping fixes based on field feedback.
- Operate in a forward-deployed, customer-facing capacity across multiple branch offices, running working sessions with field stakeholders to translate on-the-ground business needs into technical specifications and shipped features.
- Practice agentic, AI-native development with tools like Claude Code, Claude Design, and Cursor—writing prompts and agent workflows, concentrating personal effort on rigorous system design and architecture while delegating implementation to agentic coding tools, validated through comprehensive test coverage.
- Design and provision all cloud infrastructure via Infrastructure-as-Code (Bicep) on Microsoft Azure, establishing a reproducible, version-controlled deployment pipeline from the ground up.

PROJECTS & OUTSIDE EXPERIENCE

Software Engineering & Technical Community Lead

College Station, TX

Album of the Week (tamuaoiw.com) | Automation Committee

December 2023 – May 2026

- Senior developer and technical community lead over 15,000+ lines of linted, production code powering weekly operations for a member community that scaled from 25 to 180+ active members; owned architecture decisions, code review standards, and design/BI documentation.
- Designed and built autonomous, fault-tolerant background automation (Digital Ocean microservices) backed by a PostgreSQL persistence layer for weekly voting/nomination workflows, achieving 0.1% downtime through FMEA reviews and investment into end-to-end testing + staging environments.
- Built and maintained the full-stack web platform—Express.js backend, ReactJS frontend, Spotify RESTful API, and Google Identity—serving 50+ DAU; revamped CI/CD via GitHub Actions and authored a custom Dockerized testing environment, driving an estimated 20% increase in project velocity and reducing SEV-1 incidents by 4 events in Spring 2026.
- Overhauled security architecture and infrastructure: enforced JWT-authenticated endpoints via Google OAuth2 with PSQL Row-Level Security, and migrated legacy Discord bot workloads to Digital Ocean managed services, reducing outages by 73%.
- Created and presented a data-driven, Spotify-Wrapped-style year-in-review to ~100 non-technical community members—writing the narrative, building the visuals, and running the live presentation—exploring how music shaped members' semester.

Founder & Sole Developer

Dallas, TX

"2nd Brain" — Personal Knowledge Graph App

May 2026 – Present

- Architecting a local-first, personal knowledge-graph application in TypeScript for durably tracking learning over time, modeling heterogeneous sources (video, articles, PDFs, AI chat sessions) and Markdown notes as nodes within a unified, typed edge graph.
- Designed a SQLite persistence layer using a registry pattern—a central node registry for identity/universal fields with per-type tables for the rest—enforced behind a strict "save a node" API boundary.
- Built as a local Node/Bun HTTP API server with a browser-based frontend and a sparse, vim-centric editor on CodeMirror 6, a self-directed side project reflecting a personal habit of experimenting with AI-native build workflows.

EDUCATION

Texas A&M University

College Station, TX

B.S. Applied Mathematics, Computer Science Emphasis

August 2022 – May 2026

- GPA: 3.62 | Relevant Courses: (Graduate) Math Statistics | Linear Algebra II | Data Structures and Algorithms (C++)
- Coursework: Numerical Methods (iterative, non-linear, convex optimization; linear/semidefinite programming); (Grad) Optimization I & II